TRANSFINITTE · HARDWARE TRACK · PROTOTYPE BUILD GUIDE

HridayMitra: requirements, parts list and complete build guide

Monitoring only, not a diagnosis tool. Never test on a person with a pacemaker or defibrillator. Circuit values are starting values: check them in LTspice before you build.

1. What the PS requires (our checklist)

Rules	How the prototype meets them
Under 2 min a day, no all-day patch	Seated check of about 75-90 s, plus 20 s in the evening. The amulet is worn like a pendant, not stuck on.
Local language, little reading	Voice prompts, 3 lights, big buttons
Power cuts, weak internet	Battery, offline logic on the ESP32, SMS over 2G (not on Jio)
Parts under ₹5,000	See the parts list: unproven until real quotes. Owned parts must be priced in the report.

Warning sign	Measured by
Weight gain	4 load cells under the plate
Faster resting heart rate	MAX30102 finger clip (ECG optional)
Faster breathing	Amulet motion sensor at night (experimental) and symptom questions
Less activity	Amulet steps, stand-up speed from the load cells, missed check-ins
Fluid in legs	Foot-to-foot impedance at 5 kHz (50 kHz in simulation), trend against the patient's own baseline
Fluid in lungs	**Estimated** from weight, SpO2, symptoms, night sit-ups; chest belt as helper-assisted confirmation (simulation only in the prototype)
Trouble breathing lying flat	Amulet posture and night sit-ups, plus the pillow question

Deliverables: user journey, block diagram, schematic of the impedance circuit with the ESP32, parts list in INR, simulation proof (LTspice, Wokwi, Python), code with a personal baseline and a reason for every alert, a simple status screen, 200+ synthetic patients with sensitivity, specificity, median lead time and false alerts per patient-month on a patient-level split, BIDMC heart-rate test, public GitHub with setup steps, 2-page report, 5-minute pitch, the "not a diagnosis tool" disclaimer.

Judging: sensing creativity 20, ease and long-term use 20, software 20, hardware and simulation 15, science and honest testing 15, cost, India fit and pitch 10.

2. What is physical, and what is simulated

Physical (build)	Simulated or shown on screen only
Stool on a plywood plate with 4 load cells (weight); finger pulse and SpO2; 3 lights and 3 buttons; foot-to-foot impedance at 5 kHz with a resistor phantom test, then real foot plates; laptop running the alert engine; amulet with motion sensor over Bluetooth if time; SMS by SIM800L if time	Impedance at 50 kHz and the full circuit (LTspice); the chest belt; the 200+ synthetic patients and all metrics; the break-it demo; the family and ASHA message sequence (phone mock-up)

Block flow: **electrodes and sensors → ESP32 (readings, quality checks, lights, buttons) → USB serial JSON → laptop alert engine → state back to the ESP32 lights, SMS to a phone.**

3. Complete parts list

Tiers: **Must** = needed for the tier-A demo. **Should** = brownie points if time allows. **Nice** = polish. **Own** = you have it. Prices are planning ranges in ₹ (low, planning); only the load cells, HX711, SIM800L, DFPlayer, speaker and MAX30102 were web-searched, the rest are estimates.

#	Part and values	Qty	Used for	Tier	Low ₹	Plan ₹
A. Parts you already own (count their market price in the report BOM)						
1	**ESP32 WROOM-32 DevKit (classic)** Has the DAC used for the impedance signal	1	Station brain	Own	0	0
2	**AD8232 ECG module** Optional rhythm check from hand electrodes. Not in the product BOM.	1	Bonus	Own	0	0
3	**ESP32-S3 Sense** Demo stand-in for the amulet brain. Turn the camera off.	1	Amulet	Own	0	0
4	**Buttons, joystick** Use 3 push buttons: green, red, SOS	3	Answers, SOS	Own	0	0
B. Sensing and signal chain						
5	**50 kg bar load cell + HX711 board** Sold as sets. Use all four sets. Each cell has its own HX711; software adds the four weights.	4	Weight	Must	544	880
6	**MAX30102 module** I2C 0x57, 3.3 V	1	Pulse, SpO2	Must	96	150
7	**INA126 instrumentation amp, DIP-8** Single 3.3 V. Gain 10 with RG 16 kOhm. Alternatives: INA128 or INA333 breakout.	1	Reads body voltage	Must	100	200
8	**MCP6022 dual op-amp, DIP-8** Rail-to-rail, 10 MHz. Alternatives: TLV2372 (3 MHz); MCP6002 works with lower accuracy.	1	Vref and excitation buffer	Must	40	70
9	**74HC4053 analog switch, DIP-16** Selects body voltage or current-sense voltage	1	Measurement select	Must	15	30
10	**DIP sockets** 2 x 8-pin, 1 x 16-pin	3	Easy part swaps	Must	15	30
11	**Resistors, 1% metal film, 1/4 W** 10k x3, 100k x1, 4.7k x1, 220R x4 (R5 + 3 LED), 16k x1 (or 15k + 1k), 1M x2, 1k x4, 1.5k x1, 2.2k x1, 330R/680R/1k x1 each (test phantom)	~25	Impedance circuit, LEDs	Must	60	250
12	**Capacitors** 1 uF X7R ceramic or film (16 V+) x4, 100 nF x6, 10 uF x3, 10 nF x2, 100 pF x2 spare, 1000 uF 10 V electrolytic x1	~18	Coupling, decoupling, filters	Must	60	150
13	**Diodes** 1N5819 Schottky x2 (reverse-polarity protection), 1N4148 x4 spare	6	Protection	Must	10	25
14	**LEDs 5 mm: green, amber, red** With the 220R resistors above	3	Status lights	Must	10	25
C. Frame, electrodes, wiring						
15	**Stool** Any sturdy flat-top stool, seat about 40-45 cm. Foldable is optional.	1	Body	Must	250	550
16	**Plywood plate** About 40 x 40 cm, 12-18 mm thick	1	Weighing platform	Must	80	150
17	**M5 bolts, nuts, washers** M5 x 30 mm x8, nuts x8, washers x16, plus 5 mm spacers or 3 mm plywood strips	1 set	Mount load cells	Must	40	80
18	**Rubber feet** Under the free end of each load cell	4	Floor contact	Must	20	40
19	**PVC pipe 25 mm, 1 m, with flange and elbow** Post and handle bar	1 set	Handle post	Must	100	200
20	**Stainless-steel (304) strips** 4 foot plates about 8 x 5 cm and 4 hand strips about 10 x 2 cm. Round all edges.	8	Electrodes	Must	200	320
21	**Shielded 4-core cable** About 3 m, for sense lines	1	Electrode wiring	Must	40	70

22	**Hookup wire 22 AWG** Red, black, plus 2 colours, about 3 m each	1 set	Wiring	Must	40	80
23	**Project box** About 150 x 100 x 50 mm, fixed under the seat	1	Electronics enclosure	Must	60	100
24	**Breadboards, jumper wires** 2 x 830-point, M-M, M-F and F-F sets	1 set	Prototyping	Must	120	200
25	**Screws, standoffs, cable ties, heat-shrink, tape, velcro** M3 standoffs, hot-glue sticks	1 set	Assembly	Must	60	110
26	**Solder, flux** If you don't already have them	1	Soldering	Must	0	40
27	**5 V 2 A phone charger and cable** Skip if you own one	1	Power	Must	0	150
28	**Anti-slip mat** On the foot area	1	Safety	Nice	30	50
D. Messaging, voice, battery						
29	**SIM800L GSM module + antenna** 2G, needs 3.4-4.4 V and 2 A bursts. Use an Airtel, Vi or BSNL SIM (not Jio).	1	SMS and calls	Should	150	350
30	**18650 cell, holder, TP4056, MT3608 boost, slide switch** Battery, charge, 5 V rail	1 set	Battery power	Should	185	265
31	**DFPlayer Mini + microSD (2-4 GB) + 4 ohm 3 W speaker** Voice prompts. For a first demo, play voice from the laptop.	1 set	Local-language voice	Should	629	759
32	**Perfboard 7 x 9 cm + header pins** Move the circuit off the breadboard	2	Permanent circuit	Should	80	150
33	**Large 30 mm arcade buttons** Elder-friendly answer buttons	3	Answers, SOS	Nice	120	180
E. Amulet (demo)						
34	**MPU6050 breakout (or LIS2DW12)** LIS3DH chip is listed obsolete; check stock	1	Steps, posture, falls	Should	80	150
35	**LiPo 3.7 V 400 mAh with protection** Check the connector matches your board	1	Power	Should	120	180
36	**TP4056 USB-C charge module** Skip if the board already charges LiPo	1	Charging	Should	20	40
37	**Vibration motor + BC547 + 1k + 1N4148** Buzz alerts	1 set	Haptic prompt	Nice	30	50
38	**Pendant case + cotton thread** Breakaway thread, no magnets	1	Shell	Nice	60	120
39	**Pogo pins + contact board** Charging dock, no magnets	1 set	Dock	Nice	40	70
F. Test rigs (not product parts)						
40	**SG90 servo** Breathing dummy for the amulet test	1	Test	Test	100	150
41	**1 L water bottles** 5 bottles = 5 kg calibration weight	5	Calibrate weight	Test	0	0
42	**Protractor, multimeter** Angle checks and voltage checks	1 set	Test	Test	10	400

Totals to buy. Must: ₹1,960–3,900. Should: ₹1,264–1,894. Nice: ₹280–470. Must + Should: **₹3,224–5,794**. Everything: **₹3,504–6,264**. Test rigs are not counted. **For the report's product BOM, also add the market price of parts you already own** (for example the ESP32, about ₹450–600), and drop the demo-only items (breadboards, test rigs, laptop). The ₹5,000 limit is unproven until real quotes are in. If the total runs over: buy the cheaper stool, skip DFPlayer (voice from the laptop), and try a hacked cheap bathroom scale in place of the load-cell frame (saves roughly ₹400–700, unverified).

4. Pin map and wiring (classic ESP32 DevKit)

Proposed map. Check it against your board's pinout. GPIO0, 12 and 15 affect boot, so the buttons idle high on 5 and 15 and nothing pulls GPIO12 or GPIO0 low at power-up. Analog inputs use ADC1 pins so they work with Bluetooth on.

Function	ESP32 pin	Wiring notes
HX711 data x4	GPIO32, 33, 26, 27	One HX711 per load cell, powered from 3.3 V
HX711 clock (shared)	GPIO14	Same clock to all four boards
Excitation (DAC)	GPIO25	5 kHz cosine, amplitude scale 1/2
Impedance read (ADC)	GPIO35	From the INA126 output filter
Measurement select	GPIO18	74HC4053: low = body, high = current sense
I2C SDA, SCL	GPIO21, 22	MAX30102 (0x57)
ECG OUT, LO+, LO-	GPIO34, 39, 36	AD8232, optional
LEDs green, amber, red	GPIO13, 23, 2	Each through 220 Ω to GND
Buttons green, red, SOS	GPIO4, 5, 15	To GND, use internal pull-ups
SIM800L TX/RX (UART2)	GPIO17, 16	ESP32 TX → 1 kΩ → SIM800L RX with 2.2 kΩ to GND (about 2.3 V). Power the SIM800L from the battery with a 1000 µF capacitor, not from the 3.3 V pin.
DFPlayer TX	GPIO19	UART1 remapped, ESP32 TX → 1 kΩ → DFPlayer RX; DFPlayer on 5 V; speaker on its SPK pins

5. The impedance front end (the custom circuit): every value

Idea: a 5 kHz sine goes through the body via two outer electrodes. Two inner electrodes measure the voltage. A small resistor in the current path measures the current. The ESP32 reads both through the same amplifier, so **Z = R5 × (voltage across the body ÷ voltage across R5)** and the gain cancels. Bias point (Vref) is 1.65 V, half of 3.3 V.

Block	Parts and values	Connections
Vref 1.65 V	R1 10 kΩ 1%, R2 10 kΩ 1%, C1 10 µF, U1A (MCP6022) as a buffer	3V3 → R1 → node → R2 → GND; C1 from the node to GND; the node to U1A + input; U1A output tied to its - input is VREF
Excitation	C2 1 µF, R3 100 kΩ, U1B buffer	GPIO25 → C2 → node EXC_IN; R3 from EXC_IN to VREF; EXC_IN to U1B + input; U1B output is EXC_OUT (output tied to its - input)
Current path	C3 1 µF, R4 4.7 kΩ 1%, C4 1 µF, R5 220 Ω 1%	EXC_OUT → C3 → R4 → INJ_A (forefoot plate, left foot). INJ_B (forefoot plate, right foot) → C4 → node RET → R5 → VREF
Voltage sensing	R6, R7 1 kΩ (protection), R8, R9 1 MΩ (bias)	SENSE_A (heel plate, left) → R6 → SA. SENSE_B (heel plate, right) → R7 → SB. R8: SA to VREF. R9: SB to VREF
Selector 74HC4053	Two of its three switches; both select pins tied to GPIO18; third switch unused (its select to GND); enable to GND; VEE to GND; VCC 3.3 V; 100 nF	Switch X: X0 = SA, X1 = RET, common → INA + input. Switch Y: Y0 = SB, Y1 = VREF, common → INA - input. Select low = body, high = across R5
Amplifier INA126	RG 16 kΩ (15 kΩ + 1 kΩ), gain about 10, 100 nF	REF pin to VREF; V+ 3.3 V; V- GND; RG between its two RG pins; output → R11 1.5 kΩ → GPIO35, with C5 10 nF from GPIO35 to GND
Decoupling	100 nF at each IC supply pin, 10 µF on the 3.3 V rail	Close to the chips

Check pin numbers on the datasheets (INA126, MCP6022, 74HC4053) before wiring.

Expected levels (my estimates): current about 50-100 µA peak; across the body about 35-70 mV peak, so 0.35-0.7 V after the amplifier; across R5 about 11-22 mV, so 110-220 mV after the amplifier. If the R5 reading is too small, raise R5; if the body reading clips, lower the excitation scale.

Resistor-phantom test (do this before any person touches it)

1. Connect a 330 Ω resistor between INJ_A and INJ_B, with SENSE_A on the INJ_A end and SENSE_B on the INJ_B end. Run the measurement.

2. Repeat with 680 Ω and 1 kΩ. The computed Z should follow the resistor within about 5%. If it doesn't, check the select line, the bias resistors and the ADC sampling.
3. Add 1-5 kΩ in series with each outer lead to mimic poor contact. Z should barely move, because the current is measured.

Digital measurement: sample GPIO35 at 40 kHz or more (ADC continuous mode), take a Goertzel at 5 kHz over about 100 ms for each select state, and compute the amplitude ratio. If the ADC is too slow, drop the excitation to 2 kHz. Settle for about 1 s after switching the DAC on. Use the median of three readings.

LTspice proof (graph for the judges): model one leg path as outside-fluid resistance in parallel with inside-fluid resistance plus membrane capacitance; add contact resistance of 1-10 kΩ on each electrode; sweep fluid ±10% and run 5 kHz and 50 kHz. Show that the output changes with fluid and barely changes with contact resistance, and report the error.

6. Build guide, in order (with checks)

Step	Do	Done when	Time
1. Bring-up	Power the ESP32 from USB, blink the 3 LEDs, read the 3 buttons on serial	Each LED lights and each button prints	15 min
2. Weight platform	Bolt each load cell between the plywood plate and a rubber foot with spacers so the free end can flex. Wire each cell to its own HX711 (3.3 V, GND, its data pin, shared clock). Read all four and add them. Calibrate with 1, 2, 5 L water bottles (1 L is about 1 kg), store the scale factor, tare on boot.	5 kg reads 5 kg within about 0.1 kg at the center and at each corner	40 min
3. Pulse sensor	MAX30102 on I2C (3.3 V), read heart rate with a library example	Heart rate matches your wrist pulse	15 min
4. Serial protocol	Print one JSON line per check: weight, leg impedance, pulse, SpO2, button answers, quality. Receive the state (green, amber, red) from the laptop and set the LEDs.	The laptop script prints each line and a test state lights the LEDs	20 min
5. Impedance board	Build the section 5 circuit on a breadboard, one block at a time (Vref, then excitation, then selector and amplifier). Run the phantom test.	Phantom readings within about 5%	60 min
6. Frame	Fix the stool and the PVC post to the plate (bolts or flange). Fix the handle bar. Wrap hand strips on the grip (2 per hand, gap of 3 cm or more). Fix foot plates with a 2 cm gap between forefoot and heel plates. Insulate the metal from the frame. Mount the project box under the seat, cable ties for the leads.	Nothing wobbles; no metal touches another electrode	30 min
7. Electrodes live	Connect the foot plates to the impedance board and test with your own dry bare feet only after the phantom test passes and while on battery power	A stable reading; changes when you wet your feet or lift one heel	20 min
8. Integration	Run the laptop alert engine on the serial stream. Walk through green, amber and red using the break-it demo scenarios.	The LEDs follow the engine	30 min
9. Extras	Only if time is left: SIM800L SMS (AT+CMGS), DFPlayer voice, ECG on the hand electrodes, amulet	An SMS reaches a phone	as time allows

7. Firmware and data

- **Modules:** sensors (weight, impedance, pulse), quality checks (stillness, contact, signal), identity check (weight and impedance fit the patient), serial output, LED and button handler, and optional SMS and voice.
- **Daily features file** (the contract between hardware, data and engine): patient_id, day, weight_kg, leg_z_ohm, leg_z_evening_ohm, pulse_bpm, spo2, standup_s, breathless, orthopnea, took_pills, amulet_steps, amulet_situps, quality. A missing value is an empty cell, never 0.
- **Quality rules:** reject weight unless the load is steady for 5 s; reject impedance if the select-high reading is below a floor (no current flowing); reject pulse with a low-quality score. Show "no reading" instead of guessing.
- **Tare:** zero the empty stool on every power-up. Re-check with a 5 kg known weight weekly.

8. Amulet prototype (if time)

1. Wire the MPU6050 to the S3 Sense: 3.3 V, GND, SDA, SCL. Read acceleration at 25-50 Hz.
2. Compute tilt angle from gravity: lying flat, propped, upright. Count a "sit-up" as upright for more than 5 minutes between midnight and 5 am (shorten the time for the demo).
3. Count steps and movement minutes; flag worn or not worn from tiny movements.
4. Send a Bluetooth notification with angle, steps, sit-up flag and the SOS press.
5. Test with a protractor jig for angles within 5 degrees. Test breathing only with the servo breathing dummy (target: within 2 breaths a minute in at least 80% of windows); report it as experimental.

9. Safety and edge-case checklist

- **Battery power while a person touches the electrodes;** never mains. Keep current near 0.1 mA at 5 kHz and confirm against IEC 60601-1 before any real use.
- **No pacemaker or defibrillator users.** No magnets anywhere. All impedance off for implant patients.
- Round and insulate all metal edges. Keep the 18650 away from heat and metal; use a protected cell.
- Touch a grounded metal object before handling the ICs (static).
- Handle dry feet only at first. Test wet feet after the circuit is verified.

Edge case	Prototype behaviour to demo
Guest on the stool	Weight outside the patient's band: shows "Is this Ramesh-ji?" and logs a guest
Wet feet, jewellery	Impedance shifts; contact check prompts; baseline learns usual jewellery
Missing or low-quality reading	"No reading", never a guess
Missed day	Baseline kept; reminder, then SMS
Amulet forgotten	"No amulet data" flag; the stool still works
Power cut	Runs on the battery; messages queue
False alarm causes	Cause tags: infection-like, salty meal, missed medicine

10. Demo script (3 minutes), pitch and definition of done

1. **Story (20 s):** Ramaiah, 68, admitted gasping at night, weight rising for 9 days, nobody knew.
2. **Sit on the stool (40 s):** weight, pulse and leg impedance appear, green light.
3. **Break it (80 s):** inject a guest, wet feet, a missed pill, an infection-like jump, and a real worsening. Show amber then red, the spoken reason, and the SMS on a phone.
4. **Results (30 s):** the four metrics against the weight-rule baseline, with limits stated.
5. **India and business (10 s):** offline, 2G SMS, ASHA loop, kit near cost plus a monthly care-loop service (hypothesis).

Pitch (5 min): story, problem, solution, live demo, results and honest limits, India fit and business, the ask. Record a backup demo video before the final run.

Done means: block diagram, schematic, LTspice plots, code on public GitHub with setup steps, results on 200+ virtual patients, 2-page report, pitch, disclaimer, and a working tier-A stool.

11. Not verified

- All circuit values (check in LTspice), the INA126 gain and pin numbers, the 74HC4053 wiring, the ADC sampling rate and the 5 kHz measurement accuracy.
- Most prices (only six parts were web-searched), the 2G modem on your SIM, the pin map on your board.
- Amulet breathing in older people, pulse on cold or shaking fingers, and acceptance by elders.
- Whether impedance, ECG and foot-to-foot paths behave as designed on real people. Use phantoms and synthetic data for all results you report.